

24. Experimental Physics

1. Vernier Caliper:

Main Scale : Fixed

Vernier Scale : Slides



mm.

Measuring the diameter...

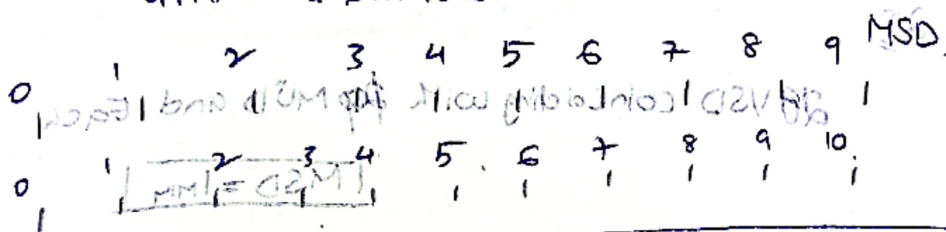
Ex: Vernier caliper.

1 MSD $\rightarrow$ 1 mm. $0.1 + 0.9 = 1.0$ mm

MSR = 0.24

VS = 0.24

$\rightarrow$ 10 Vernier Scale Divisions are coinciding with a Main Scale Divisions.



"Least Count" = $\frac{1 \text{ MSD}}{10} = \frac{1 \text{ mm}}{10} = 0.1 \text{ mm}$

MSR $\rightarrow$ 0.24

VS $\rightarrow$ 0.24

LC = 1 MSD - 1 VSD

10 VSD $\rightarrow$ 9 mm
1 VSD $\rightarrow$ 0.9 mm

LC = 1 mm - 0.9 mm

LC = 0.1 mm

Reading = MSR + (LC)(VSR)

MSR = 0.24

VS = 0.24

where.

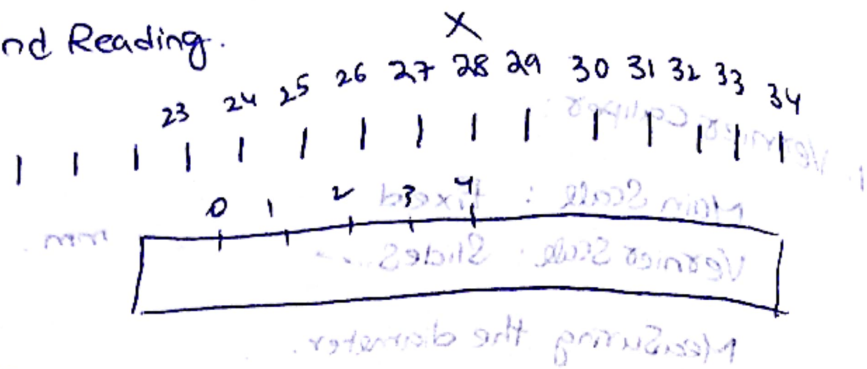
MSR $\rightarrow$ Main Scale Reading

LC $\rightarrow$ Least Count

VSR $\rightarrow$ Vernier Scale Reading

0.1 mm

1Q. Find Reading.



$$MSR = 23$$

$$VSR = 4$$

$$\text{Reading} = 23 + (0.1)(4) = 23.4 \text{ mm}$$

2Q.

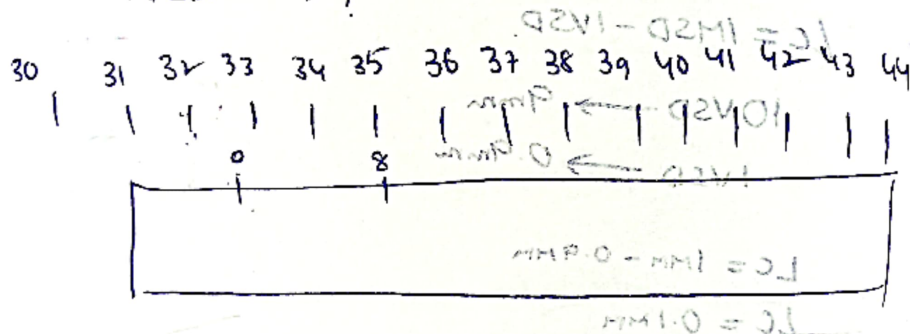
20 VSD coinciding with 10 MSD and each

$$1 \text{ MSD} = 1 \text{ mm}$$

$$20 \text{ VSD} \rightarrow 19 \text{ mm}$$

$$X = \frac{19}{20} = \frac{19}{20} \times 10^{-1} = 0.95$$

$$1 \text{ VSD} \rightarrow 1$$



$$MSD = 32$$

$$VSR = 8$$

$$LC = MSD - VSR$$

$$LC = 0.05 \text{ mm}$$

$$\text{Reading} = 32 + (0.05)8$$

$$= 32.40$$

Screw Gauge:

$$LC = \frac{\text{Pitch}}{\text{NO. OF Circular Scale Division}}$$

NO. OF Circular Scale Division

$P \rightarrow 1\text{mm}$
 $CSD \rightarrow 100$

$$LC = \frac{1}{100} = 0.01\text{mm}$$

$$\text{Reading} = \text{MSR} + LC(\text{CSR})$$

Ex: $\text{MSR} = 6$

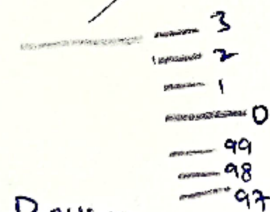
$\text{CSR} = 75$

$$\text{Reading} = 6 + (0.01)(75)$$

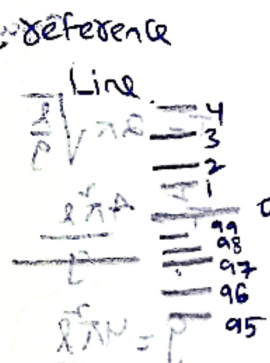
$$\text{Reading} = 6.75\text{mm}$$



No Zero Error.



Positive Zero Error...



Negative Zero Error.

$$LC = 0.01\text{mm}$$

$\text{Zero} = 3(0.01)$
 $\text{Error} = 0.03\text{mm}$

$\text{Zero} = -0.03$

$\text{O.R} = 26.38\text{mm}$
 $\text{A.R} = 26.41\text{mm}$

i. Observed Reading: 76.24 mm

ii. Actual Reading: 76.21 mm

EXPERIMENTAL PHYSICS:

: Measurement.

: Calculate.

Example: Resonance tube Exp.

Aim: To determine the Speed of Light.

of Sound in Air.

→ Measure l_1 & l_2

→ Calculate $v = 2f(l_2 - l_1)$

Example 2: Simple Pendulum Exp.

Aim: To determine the Value of g .

Measure l & T .

Calculate $g = \frac{4\pi^2 l}{T^2}$

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$$T^2 = \frac{4\pi^2 l}{g}$$

$$g = \frac{4\pi^2 l}{T^2}$$

$$50.0 - 0.0 = 50.0$$

$$10.0 \pm 0.1$$

$$10.0 \pm 0.1$$

$$(10.0) \pm 0.1$$

$$10.0 \pm 0.1$$

$$10.0 \pm 0.1$$

Observed Reading: 10.0 mm

Actual Reading: 10.0 mm